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(54) **HANDGUN BRACE/STOCK ADAPTOR**

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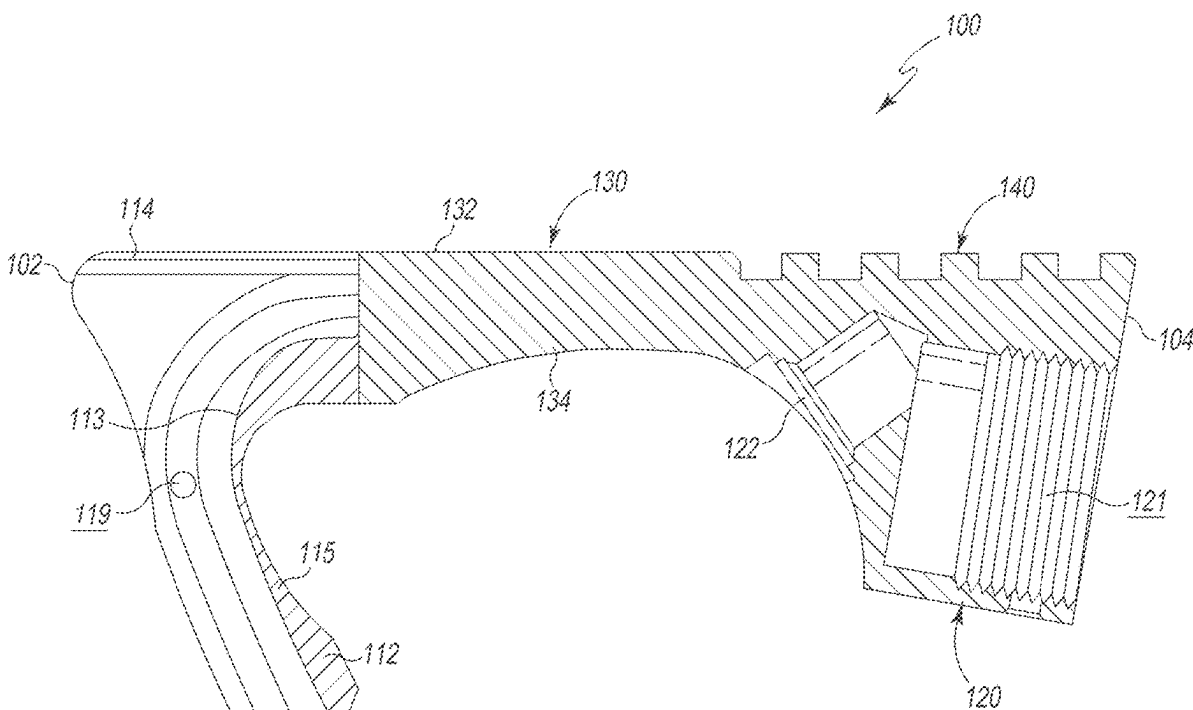
(57) **ABSTRACT**

(22) Filed: **Sep. 27, 2024**

The pistol/stock adaptor allow conventional AR-style rifle stocks and pistol braces to be attached to the grip of a conventional handgun. The adaptor is a single integral component having a grip saddle that affixes the adaptor to a handgun, a buffer tube receiver that connects the buffer to the adaptor, and an elongated neck or bridge integrally connecting the grip saddle and the buffer tube receiver. The adaptor allows a shooter to “shoulder” the handgun against the shooter’s torso, while maintaining a conventional two handed grip on the handgun.

Related U.S. Application Data

(60) Provisional application No. 63/556,611, filed on Feb. 22, 2024.



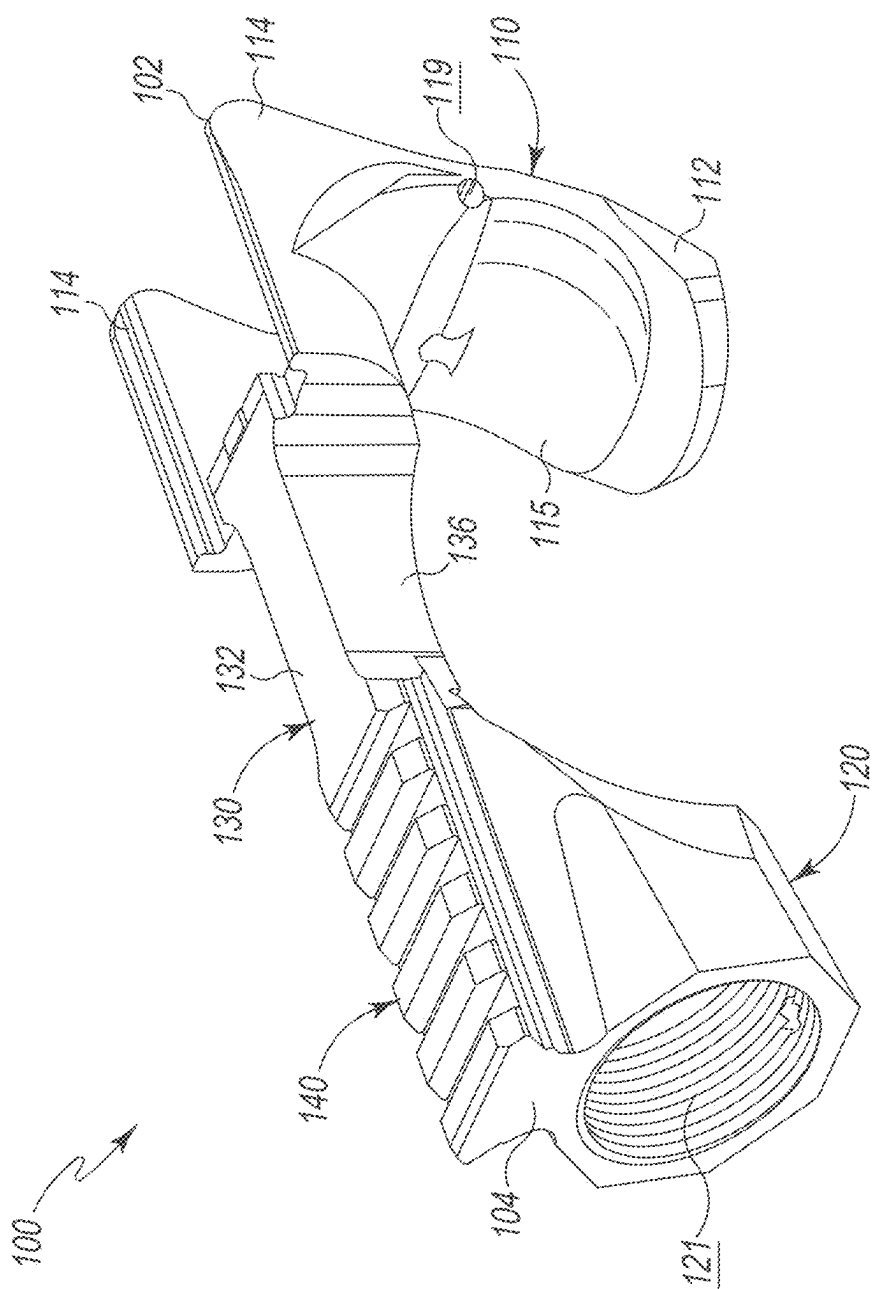
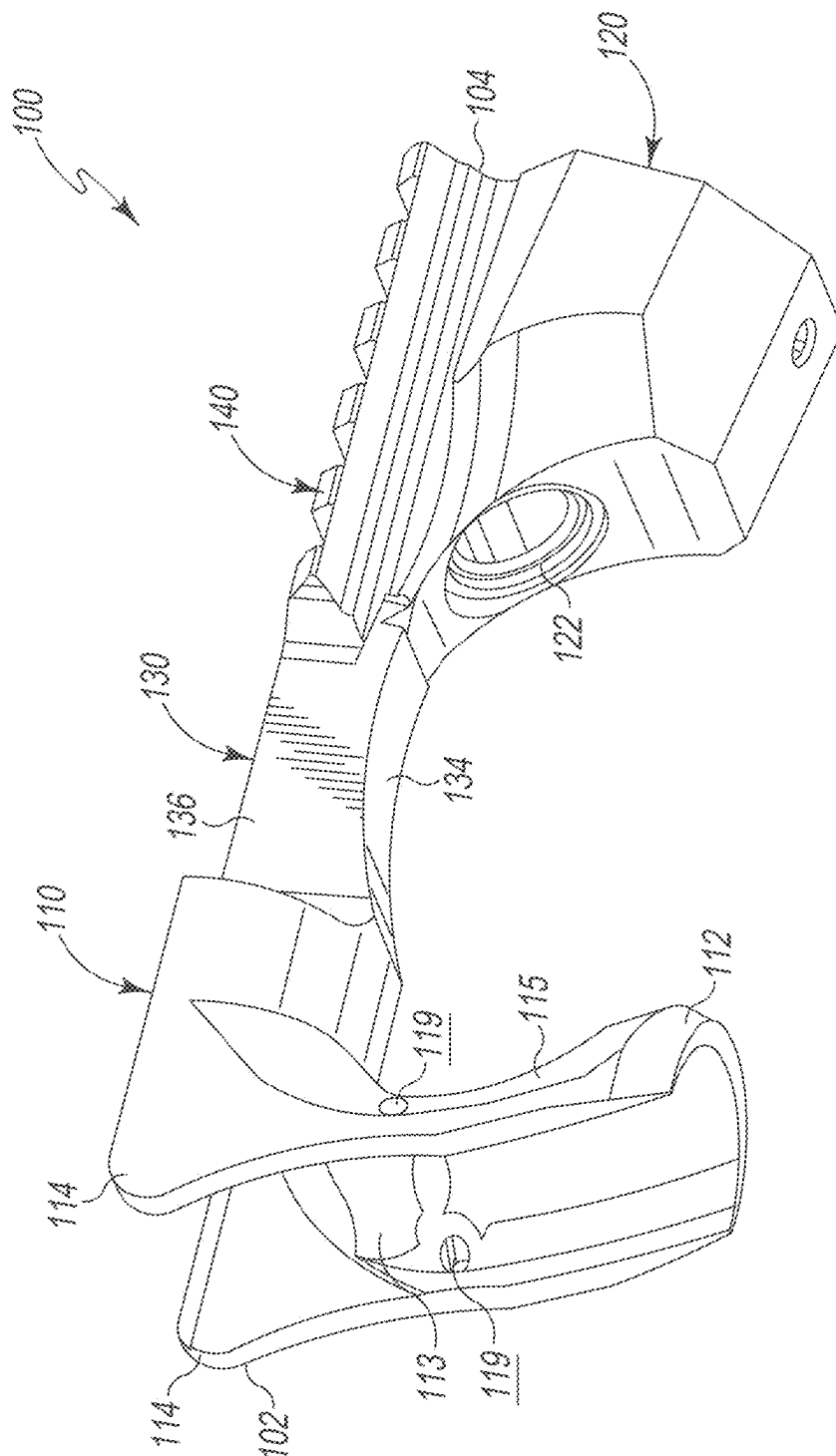


Fig. 1



Lib. 2

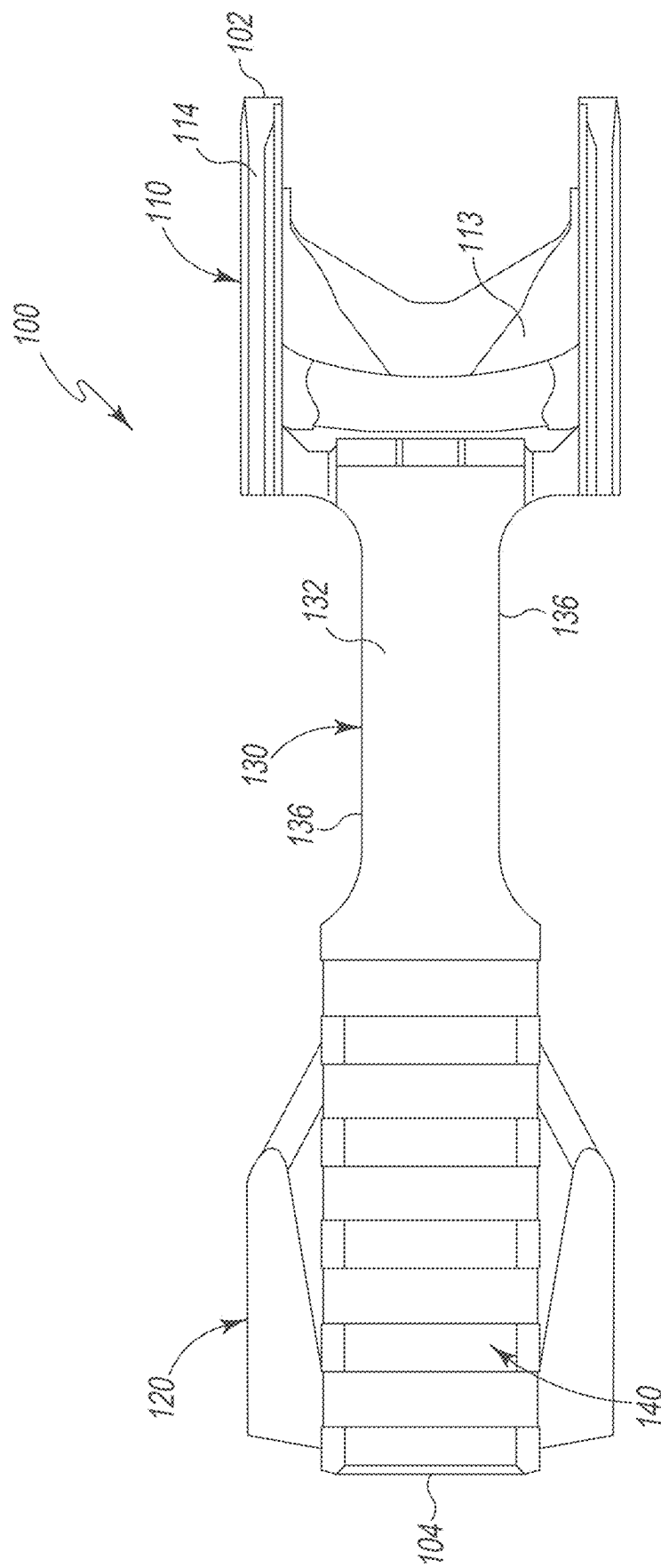


Fig. 3

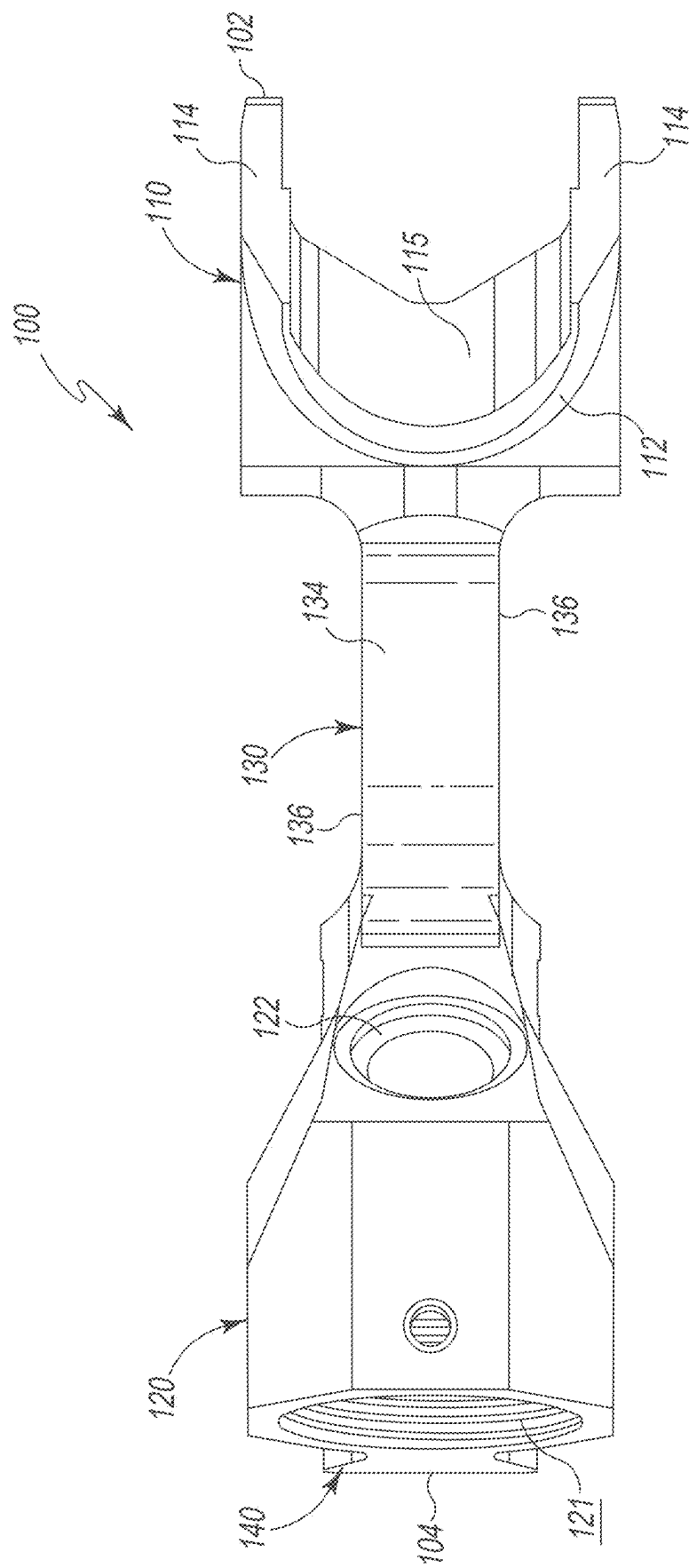


Fig. 4

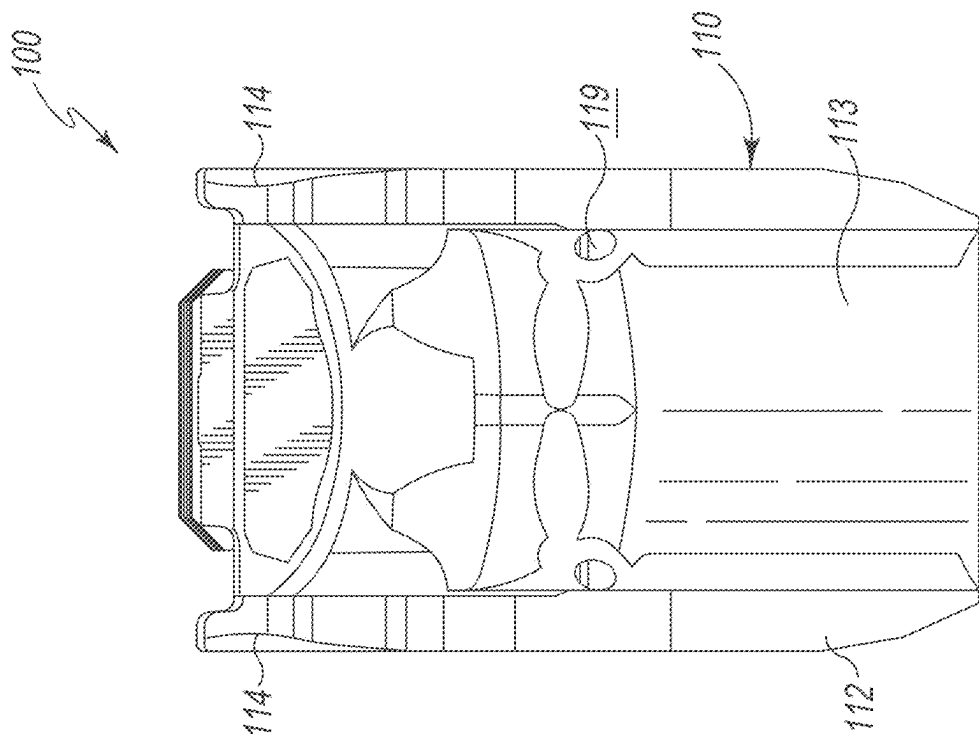


Fig. 5

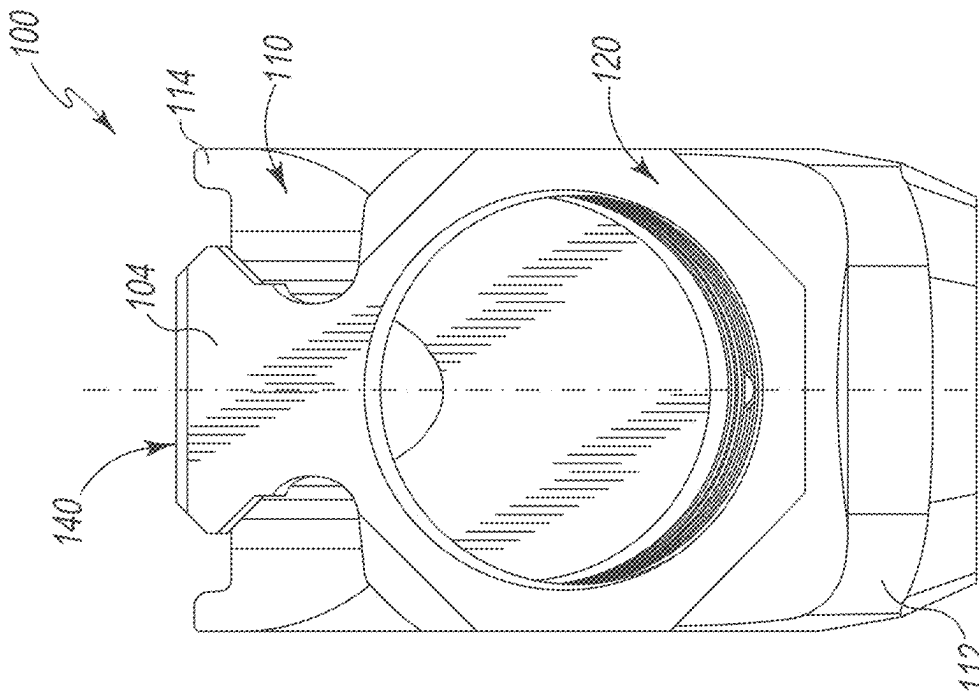


Fig. 6

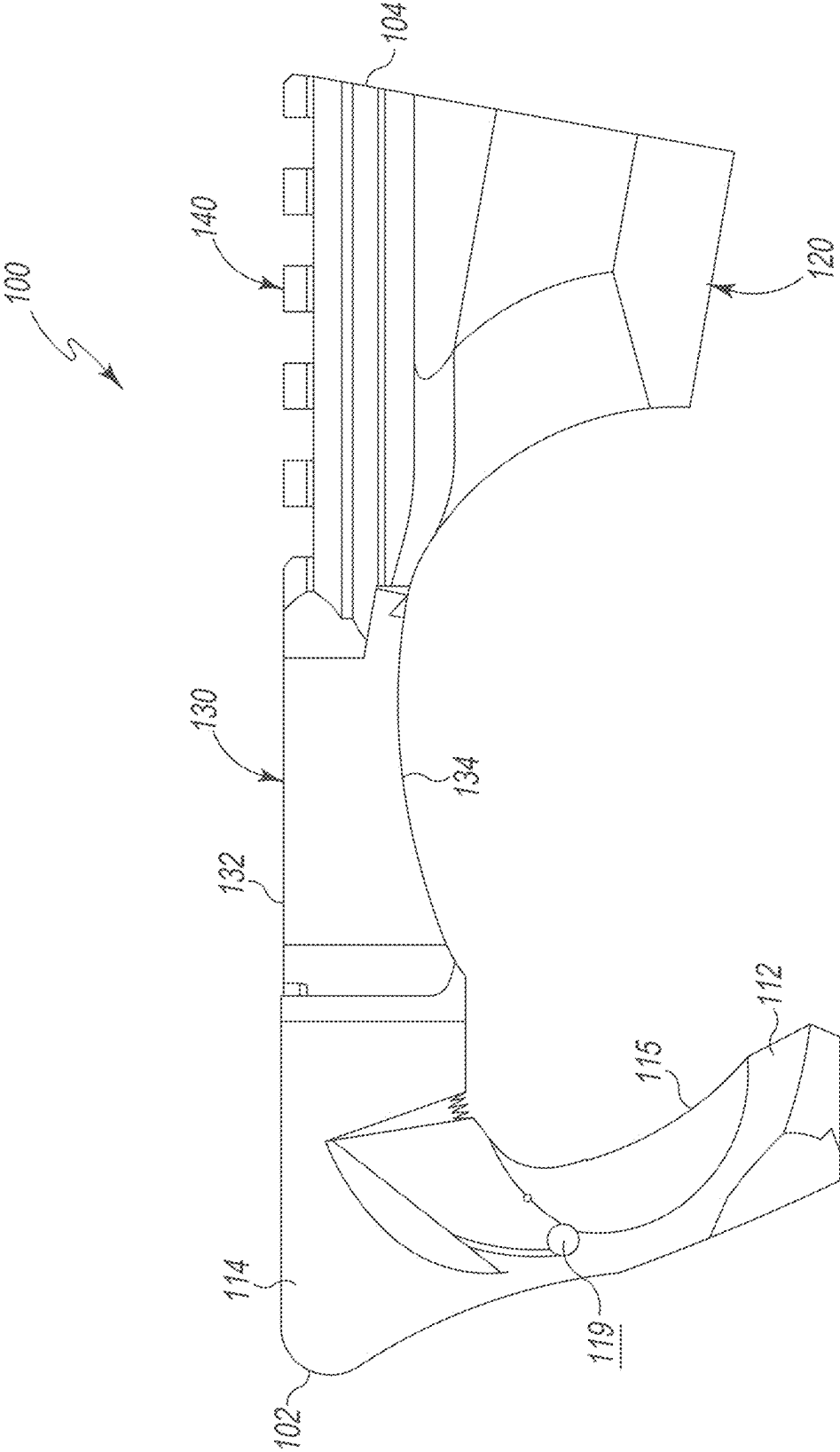


Fig. 7

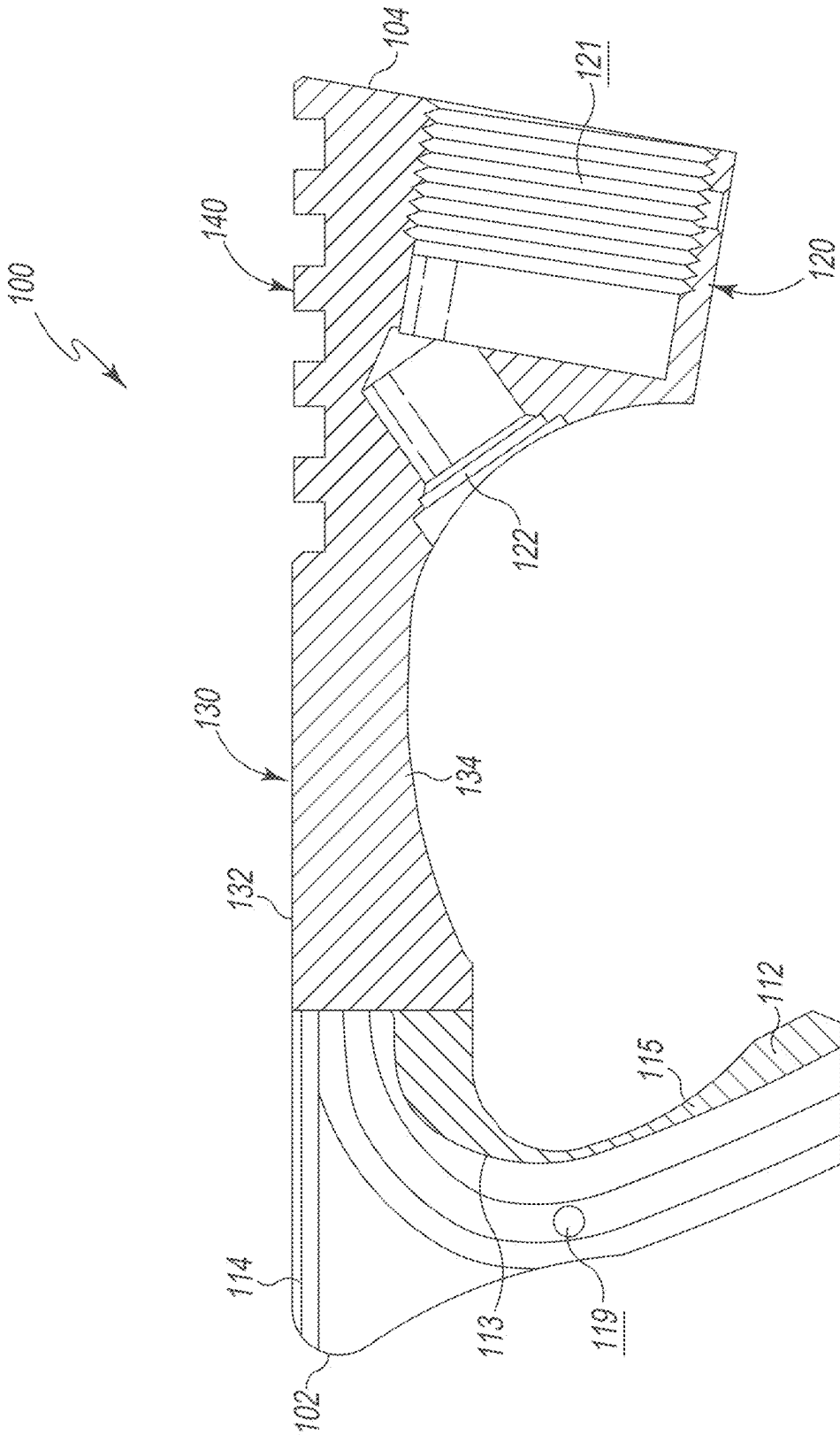


Fig. 8

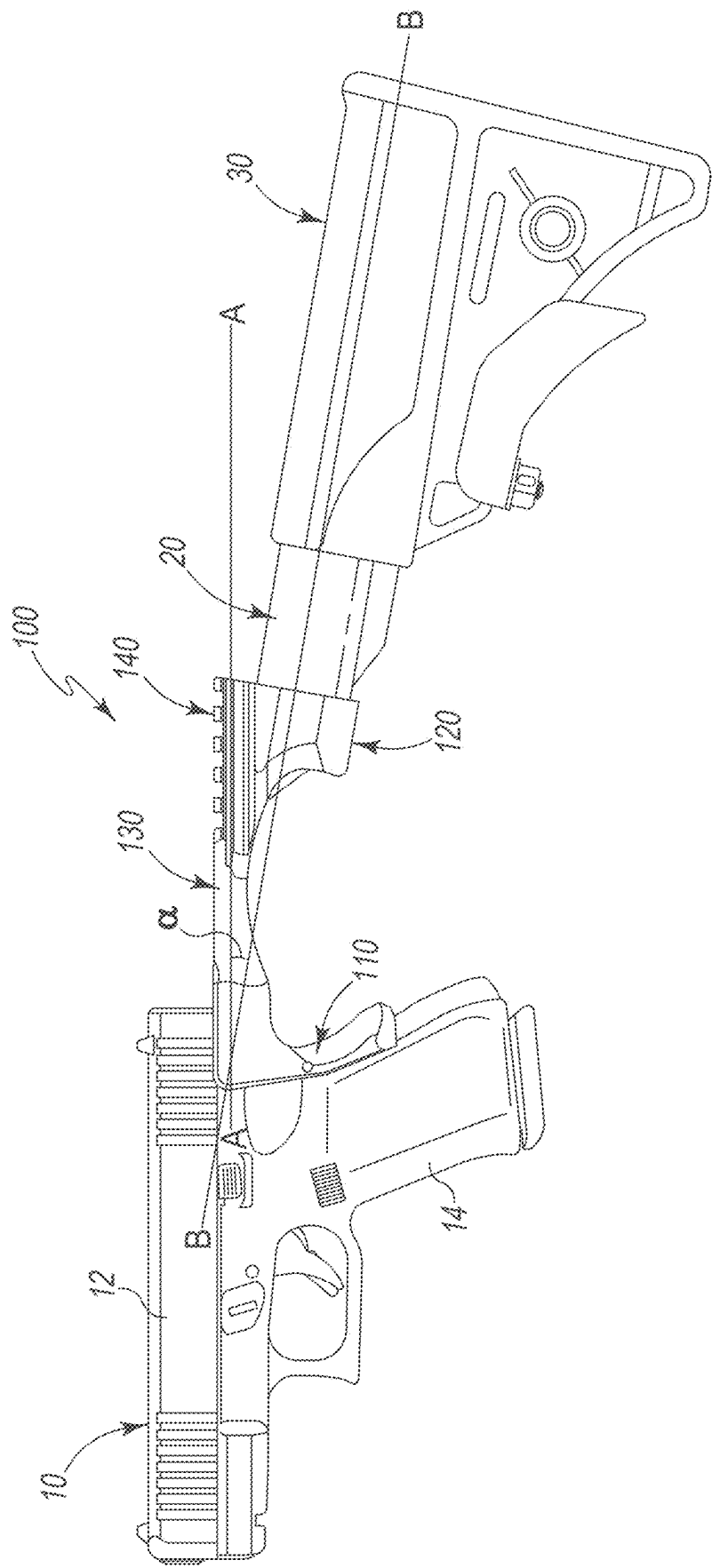


Fig. 9

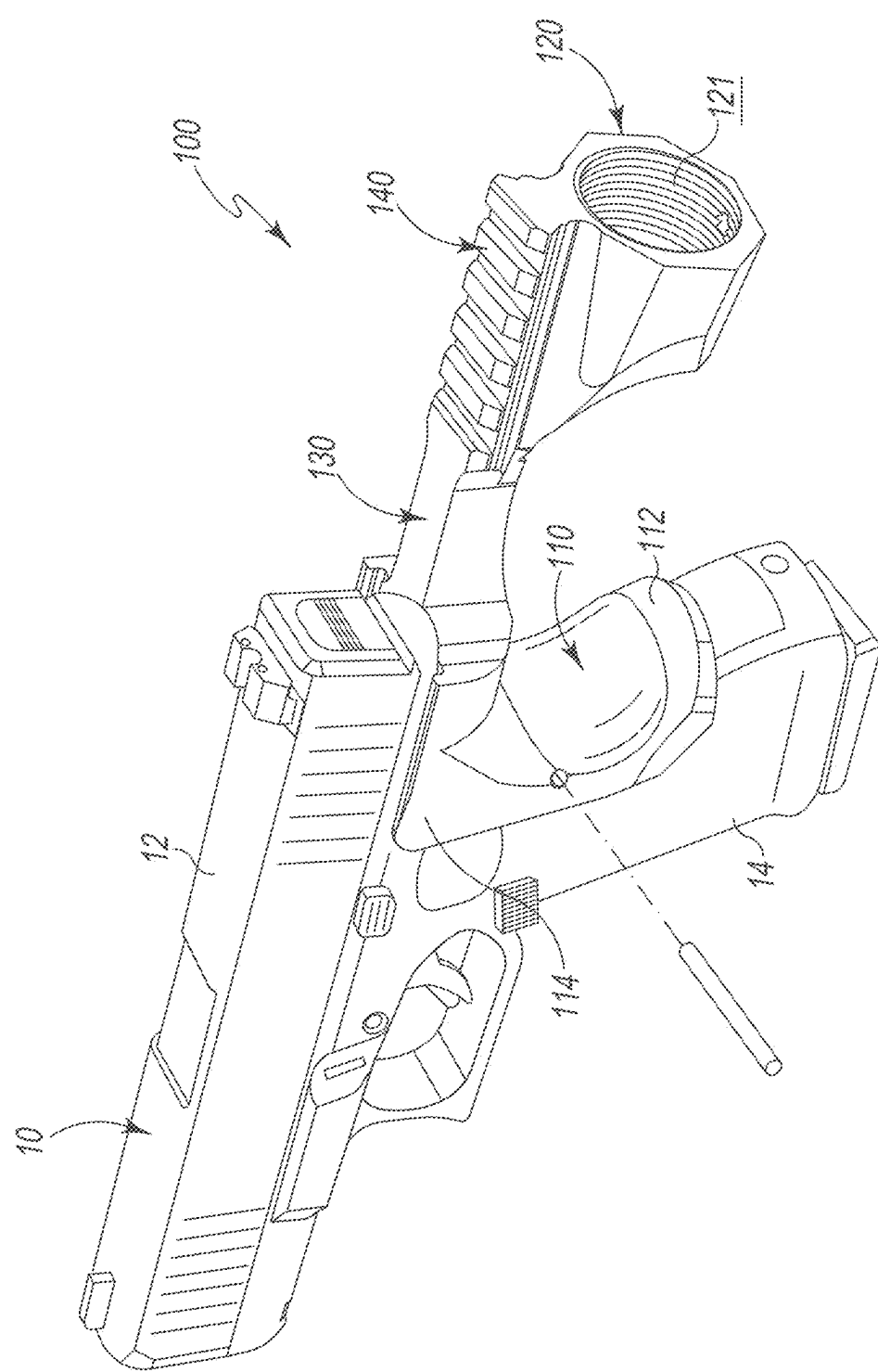


Fig. 10

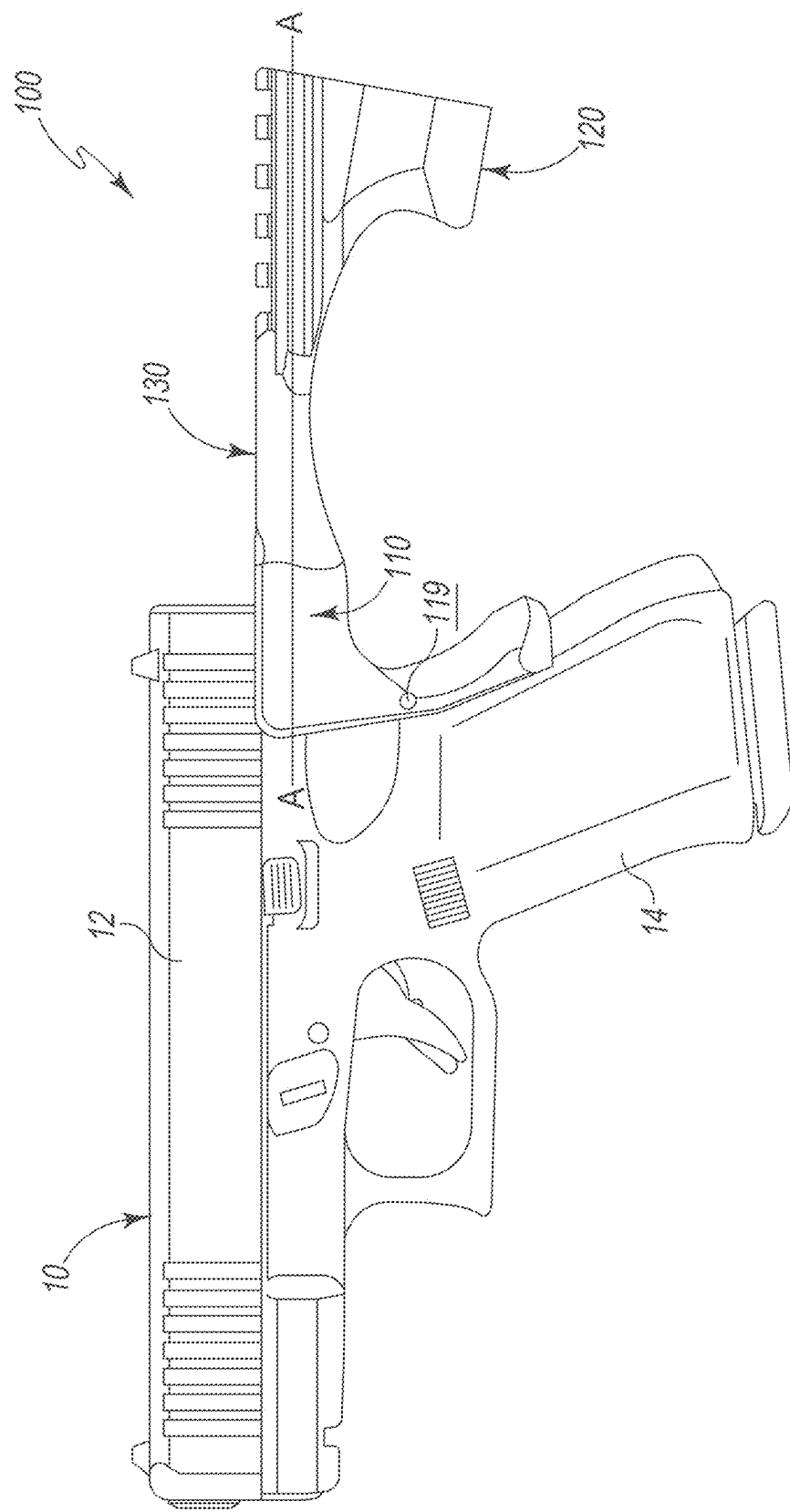


Fig. 11

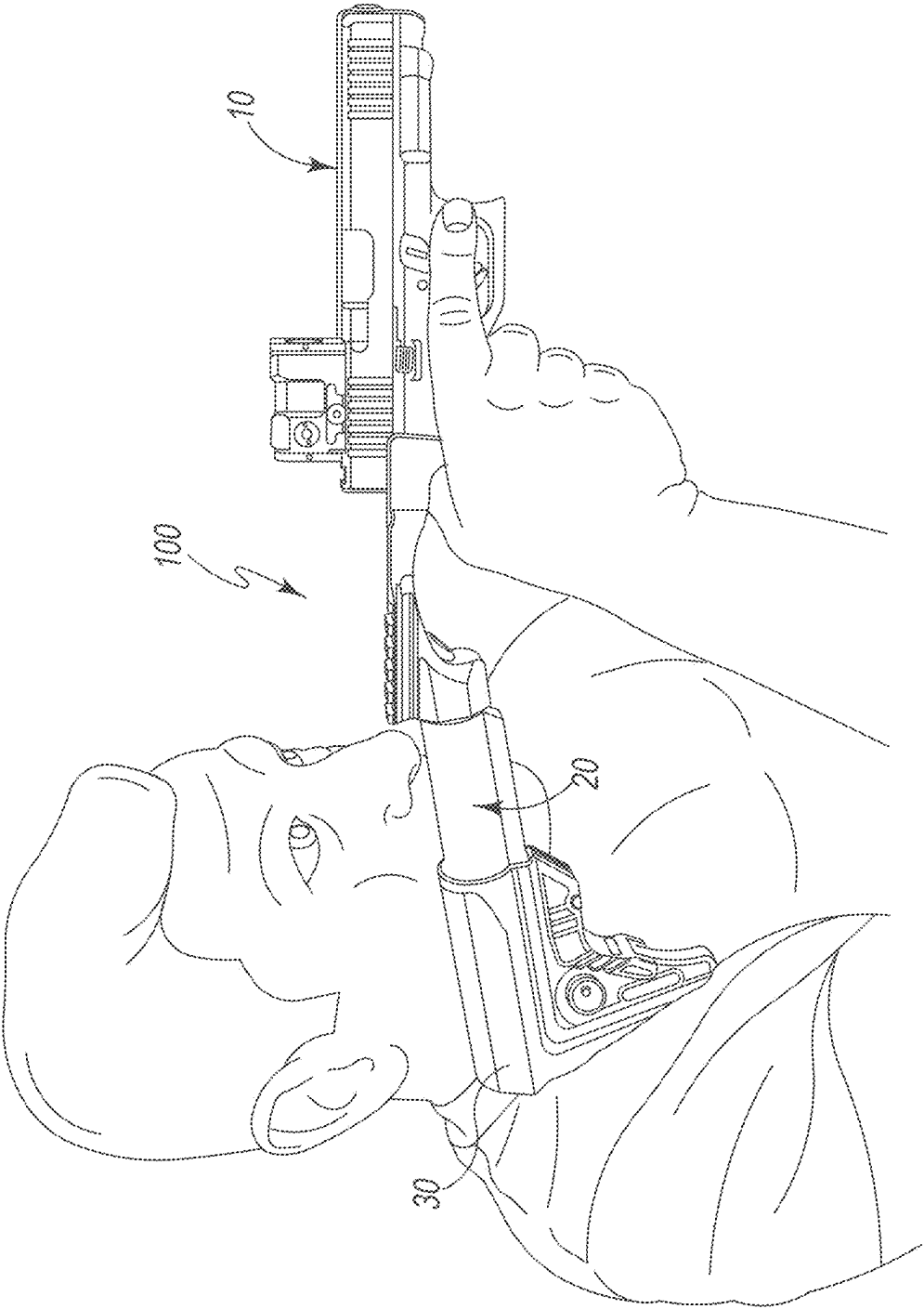


Fig. 12

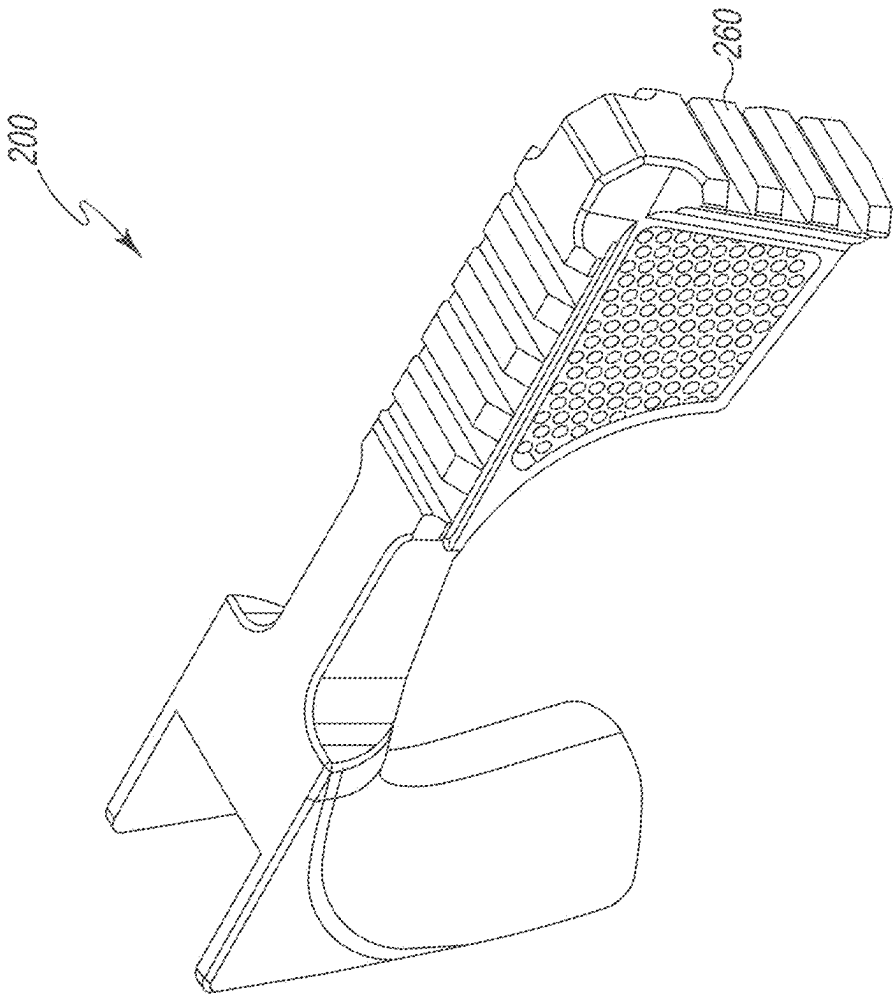


Fig. 13

HANDGUN BRACE/STOCK ADAPTOR

[0001] This application claims the benefit of U.S. Provisional Patent Application, Ser. No. 63/556,611 filed Feb. 22, 2024, and U.S. Provisional Patent Application, Ser. No. 63/568,326 filed Mar. 21, 2024, the disclosures of both provisional patent applications are hereby incorporated by reference.

[0002] This invention relates to firearm attachments, and has specific relevance to firearm attachments that connect a rifle stock or brace to a pistol frame.

BACKGROUND AND SUMMARY OF THE INVENTION

[0003] Handguns, such as pistols and revolvers, are traditionally designed and intended for one or two handed manipulation. By contrast, long guns, such as rifles and shotguns, are “shouldered” with both hands and cheek on the weapon and the weapon stabilized or rested against the shooter’s shoulder. Shouldering a firearm provides at least three points of contact to stabilize the weapon while a handgun has only one or two points of contact, thereby making handguns less stable and effective under certain circumstances.

[0004] The handgun brace/stock adaptors of this invention allows commonly available AR-style rifle stocks and pistol braces to be attached to the grip of a conventional handgun. The adaptors are a single integral component having a grip saddle that affixes the adaptor to a handgun, a buffer tube receiver that connects the buffer to the adaptor, and an elongated neck or bridge integrally connecting the grip saddle and the buffer tube receiver. The adaptors allow a shooter to “shoulder” the handgun against the shooter’s torso, while maintaining a conventional two handed grip on the handgun, which improves accuracy and safety. When shouldered, the attachment positions the bore axis of the handgun close to the shooter’s natural line of sight. The adaptor also integrates an accessory rail interface and a quick detach feature for mounting optics and connecting a rifle sling to the attachment. The adaptors of this invention may be modified and configured in various embodiments for use with any conventional handgun, but are specifically designed and intended for semi-automatic pistols. Similarly, the handgun attachment of this invention may be adapted and configured to use with any style of rifle stock, but is specifically designed and intended for the various commonly available AR-15 style rifle stocks and braces.

[0005] The above described features and advantages of the handgun brace/stock adaptor of this invention, as well as others, will become more readily apparent to those of ordinary skill in the art by reference to the following detailed description and accompanying drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] The present invention may take form in various system and method components and arrangement of system and method components. The drawings are only for purposes of illustrating exemplary embodiments and are not to be construed as limiting the invention. The drawings illustrate the present invention, in which:

[0007] FIG. 1 is a top rear perspective view of an exemplary embodiment of the handgun brace/stock adaptor of this invention:

[0008] FIG. 2 is a bottom front perspective view of the adaptor of FIG. 1;

[0009] FIG. 3 is a top view of the adaptor of FIG. 1;

[0010] FIG. 4 is a bottom view of the adaptor for FIG. 1;

[0011] FIG. 5 is a rear view of the adaptor of FIG. 1;

[0012] FIG. 6 is a front view of the adaptor of FIG. 1;

[0013] FIG. 7 is a side view of the adaptor of FIG. 1;

[0014] FIG. 8 is a side sectional view of the adaptor of FIG. 1;

[0015] FIG. 9 is a side view of the adaptor of FIG. 1 connecting a conventional pistol to a conventional AR style buffer tube and collapsible stock;

[0016] FIG. 10 is a side view of the adaptor and pistol of FIGS. 9; and

[0017] FIG. 11 is a top rear perspective view of the adaptor and pistol of FIG. 9;

[0018] FIG. 12 is a side view of a shooter shouldering the pistol, adaptor, buffer and stock of FIGS. 9; and FIG. 13 is a rear perspective view of a second exemplary embodiment of the adaptor of this invention.

DESCRIPTION OF THE PREFERRED EMBODIMENT

[0019] In the following detailed description of the preferred embodiments, reference is made to the accompanying drawings that form a part hereof, and in which is shown by way of illustration specific preferred embodiments in which the invention may be practiced. These embodiments are described in sufficient detail to enable those skilled in the art to practice the invention, and it is understood that other embodiments may be utilized and that logical, structural, mechanical, electrical, and chemical changes may be made without departing from the spirit or scope of the invention. To avoid detail not necessary to enable those skilled in the art to practice the invention, the description may omit certain information known to those skilled in the art. The following detailed description is, therefore, not to be taken in a limiting sense, and the scope of the present invention is defined only by the appended claims.

[0020] The handgun brace/stock adaptor of this invention (the “adaptor”) allows conventional AR-style rifle stocks and pistol braces to be attached to the grip of a conventional handgun. Like conventional handguns, AR-15 style stocks and braces, and AR-15 style buffer tubes are well known in the firearms industry and readily available to both the military/law enforcement and civilian markets. The adaptors of this invention may be adapted and configured in various embodiments for use with any conventional handgun, but are specifically designed and intended for semi-automatic pistols. Similarly, the handgun attachment of this invention may be adapted and configured to use with any style of rifle stock, but is specifically designed and intended for the various AR-15 style rifle stock or pistol brace. The adaptor allows a shooter to “shoulder” the handgun against the shooter’s torso, while maintaining a conventional two handed grip on the handgun. When shouldered, the attachment positions the bore axis of the handgun close to the shooter’s natural line of sight. The adaptor also integrate an accessory rail interface and a quick detach feature for mounting optics and connecting a rifle sling to the attachment.

[0021] FIGS. 1-12 show an exemplary embodiment of the handgun brace/stock adaptor of this invention, which is designated generally as reference number 100. For simplic-

ity of illustration and explanation, adaptor **100** is illustrated connecting a conventional semi-automatic pistol **10** to a conventional AR-15 style buffer tube **20** and collapsible rifle stock **30**.

[0022] Pistol **10** is illustrated as a conventional striker fired pistol, such as the Glock **19**, manufactured by the Austrian company, Glock Ges.m.b.H. The construction and operation of semi-automatic pistols are generally well known and understood in the art. Pistol **10** includes a barrel (not shown), a reciprocating slide **12** and a polymer frame, having a grip **14**. Pistol grip **14** has removable roll pin (“the trigger housing pin”) **16**. Roll pin **16** forms part of the fire control mechanism of striker fired pistols. Pin **16** is pressed into a lateral through bore in grip **14**. As shown, pistol **10** also has a protrusion (“beaver tail”) **18** extending from the back of grip **14**. Beaver tail **18** extends over the top of a shooter’s hand and is designed to prevent the user’s hand from slipping off the grip of the gun during recoil.

[0023] Buffer tube **20** and rifle stock **30** are illustrated as generic conventional AR-15 style buffer tubes and stocks. Buffer tube **20** has an externally threaded pistol end **22**. Rifle stock **30** slides onto buffer tube **20** and can be selectively positioned along the length of the buffer tube. Generally, rifle stocks and pistol braces are slidably mounted over the buffer tube and secured at different positions along the tube to selectively set the weapon’s “length of pull.” In AR-15 style weapons, buffer tubes are connected to the weapon’s upper receiver and houses the weighted buffers and buffer springs, which dampen the movement of the bolt carrier group and form part of the AR-15’s fire control mechanism.

[0024] As shown, handgun brace/stock adaptor **100** is cast or machined from an aluminum alloy as a single integral component. In other exemplary embodiments, adaptor **100** may be formed or molded from a suitable polymer material. As shown, adaptor **100** has a grip saddle **110**, a buffer tube receiver **120** and a elongated neck or bridge integrally connecting the grip saddle and buffer receiver. Generally, adaptor is an elongated element with a grip end **102** and a stock end **104** and a longitudinal axis A-A that is generally parallel to the bore axis of pistol **10** and slide **12** when the attachment is affixed to pistol **10**.

[0025] Grip saddle **110** forms part of pistol end **102** of adaptor **100** and is configured to receive and mate against pistol grip **14** and beaver tail **18** of pistol **10** (FIGS. 8-10). As shown, grip saddle **110** attaches to pistol **10** along the rear of pistol grip **14** beneath slide **12** partially covering the upper rear portion of pistol grip **14** and enclosing “beaver tail” **18**. Grip saddle **110** has contoured grip cover **112** and a pair of side members or “wings” **114**. Grip saddle **110** also has a curved inner surface **113** configured to correspond to the shape and contour of pistol grip **14** and beaver tail **18**. Side wings **114** extend over and around beaver tail **18** beneath slide **12**. Grip saddle **110** is affixed to pistol grip **14** by pin **16** which extends through aligned bores **119** in grip saddle **110** as well as its intended lateral bore in pistol grip **14**. Often, roll pin **16** is a longer replacement pin from the trigger housing pin normally used in pistol **10**. Roll pin **16** provides a mechanical component that affixes adaptor **100** to pistol **10**. Adhesives and adhesive tapes (not shown) may be applied to the mating surfaces of grip saddle **110** and pistol grip **14** to further secure adaptor **100** to pistol **10**. Grip saddle **110** provides a single point of connection to pistol **10** at the

top of grip **14** without any addition support beam at the bottom of the grip, which may interfere with the shooter’s manipulation of the pistol.

[0026] Buffer tube receiver **120** forms part of stock end **104** of adaptor **100** and is configured to received buffer tube **20** (FIGS. 9 and 12). The rear of buffer tube receiver **120** has an internally threaded cylindrical bore **121** for receiving buffer tube **20**. The center axis B-B of buffer tube bore **121** extends at an angle α from attachment longitudinal axis A-A. Angle α range generally between 10 degrees and 35 degrees. Angle α is selected to elevate pistol **10** in the shooter’s field of view when pistol **10** is shouldered using adaptor **100**. The front of buffer tube receiver **120** has a swivel socket **122** for receiving a conventional quick detach sling swivel (not shown). Socket **122** allows a rifle sling to be easily connected to adaptor **100** using a quick detach swivel. The top of buffer tube receiver **120** has an integral rail interface **140** for detachably connecting firearm accessories, namely optics and optic mounts (not shown) to adaptor **100**. For simplicity of explanation, rail interface **140** is illustrated as a Mil-spec **1913** rail (“Picatinny rail”) although other rail interfaces, such as the M-LOK® system from Magpul Industries Corporation of Austin, Texas, may be integrated into buffer tube receiver **120**.

[0027] Bridge **130** is a narrow beam that integrally connects grip saddle **110** and buffer tube receiver **120**. Bridge **130** has a generally rectangular cross section. Bridge **130** has a flat top surface **132**, a bottom surface **134** and opposed sides **136**. As shown the width of bridge **130** is narrower than the respective widths of grip saddle **110** and buffer tube receiver **120**. As shown, the top edges of grip saddle **110**, bridge top surface **132** and the top of rail interface **140** all lie in a plane parallel axis A-A and the bore axis of pistol **10**. The alignment of grip saddle **110**, bridge **130** and rail interface **140** allow pistol slide **12** to cycle unimpeded and reciprocate adaptor **100** when affixed to pistol **10**.

[0028] As shown in FIGS. 12, adaptor **100** is configured so that a shooter can easily grasp pistol **10** and grip saddle **110** with either hand when the adaptor is affixed to pistol **10**. Adaptor **100** allows a shooter to support or “shoulder” pistol **10** similar to a long gun, while maintaining a two handed grip of the pistol. The configuration adaptor **100** also allows a shooter to present and holster pistol **10** with the buffer tube and stock connected to the pistol without altering the shooter’s normal weapon manipulation techniques. Pistol **10** can also be holstered with adaptor **100**, buffer tube **20** and stock **30** still attached to the pistol.

[0029] FIG. 13 illustrates a second exemplary embodiment of the adaptor of this invention, which is designated generally as reference numeral **200**. Adaptor **200** is similar in form, function and construction to adaptor **100**, but is intended for uses with “Picatinny” style rifle stocks, such as the ones available from Midwest Industries, Inc. of Waukeasha, Wisconsin. Adaptor **200** has a second rail interface **260** that accepts a conventional Picatinny style rifle stock, brace or pad (not shown) instead of a buffer tube. Rail interface **260** is integrated into the end wall of adaptor **200**.

[0030] It should be apparent from the foregoing that an invention having significant advantages has been provided. While the invention is shown in only a few of its forms, it is not just limited but is susceptible to various changes and modifications without departing from the spirit thereof. The embodiment of the present invention herein described and illustrated is not intended to be exhaustive or to limit the

invention to the precise form disclosed. It is presented to explain the invention so that others skilled in the art might utilize its teachings. The embodiment of the present invention may be modified within the scope of the following claims.

I claim:

1. An adaptor for connecting an AR-style buffer tube to a pistol, where the pistol includes a pistol frame having a pistol grip and reciprocating slide, and where the buffer tube has a first tube end and a second tube end, the adaptor comprising:

an elongated adaptor body having a first adaptor body end thereof and a second adaptor body end thereof,

the adaptor body has a grip saddle at the first adaptor body end configured to receive the pistol grip and securely affixed the adaptor to the pistol grip, so that the pistol slide reciprocates over the adaptor body,

the adaptor body has a buffer tube receiver at the second adaptor body end configured to receive the first tube end of the buffer tube and affix the buffer tube to the adaptor,

the adaptor body has a bridge part integrally extending axially between the grip saddle and the buffer tube receiver.

2. The adaptor of claim 1 wherein the grip saddle includes a grip cover that extends over the pistol grip and is contoured to allow the grip saddle to be grasped by a shooter.

3. The adaptor of claim 1 wherein the grip saddle also includes a pair of side members that extend over and around the sides of the pistol grip.

4. The adaptor of claim 3 wherein the adaptor body is secured to the pistol grip by a pin that extends laterally through the pistol grip between the pair of side members of the grip saddle.

5. The adaptor of claim 1 wherein the adaptor body is secured to the pistol grip by an adhesive applied between the grip saddle and the pistol grip.

6. The adaptor of claim 1 wherein the buffer tube receiver has a cylindrical bore for axially receiving the first buffer tube end so that the buffer tube extends longitudinally from the adaptor body at the second adaptor body end.

7. The adaptor of claim 6 wherein the adaptor body has a longitudinal axis thereof, when the adaptor is connected to the pistol, the longitudinal axis of the adaptor body is parallel to the pistol slide, the buffer tube extends from the adaptor body at an angle relative to the longitudinal axis of the adaptor body.

8. The adaptor of claim 1 wherein the buffer tube receiver includes a swivel socket.

9. The adaptor of claim 1 wherein the buffer tube receiver includes a rail interface for detachably connecting firearm accessories to the adaptor.

10. The adaptor of claim 9 wherein the buffer tube receiver includes a second rail interface for detachably connecting the firearm stock to the adaptor.

11. An adaptor for connecting a rifle stock to a handgun, where the handgun has barrel and a handgun grip, and where the rifle stock includes an elongated buffer tube a first tube end and a second tube end and a stock part connecting the second tube end buffer tube, the adaptor comprising:

an elongated adaptor body having a first adaptor body end thereof and a second adaptor body end thereof,

the adaptor body has a grip saddle at the first adaptor body end configured to receive the handgun grip and securely affix the adaptor to the handgun grip,

the adaptor body has a buffer tube receiver at the second adaptor body end configured to receive the first tube end of the buffer tube and affix the buffer tube to the adaptor,

the adaptor body has a bridge part integrally extending axially between the grip saddle and the buffer tube receiver.

12. The adaptor of claim 11 wherein the grip saddle includes a grip cover that extends over the handgun grip and is contoured to allow the grip saddle to be grasped by a shooter.

13. The adaptor of claim 11 wherein the grip saddle also includes a pair of side members that extend over and around the sides of the handgun grip.

14. The adaptor of claim 13 wherein the adaptor body is secured to the handgun grip by a pin that extends laterally through the handgun grip between the pair of side members of the grip saddle.

15. The adaptor of claim 11 wherein the adaptor body is secured to the handgun grip by an adhesive applied between the grip saddle and the handgun grip.

16. The adaptor of claim 11 wherein the buffer tube receiver has a cylindrical bore for axially receiving the first buffer tube end so that the buffer tube extends longitudinally from the adaptor body at the second adaptor body end.

17. The adaptor of claim 16 wherein the adaptor body has a longitudinal axis thereof, such that when the adaptor is connected to the handgun, the longitudinal axis of the adaptor body is parallel to the handgun barrel, the buffer tube extends from the adaptor body at an angle relative to the longitudinal axis of the adaptor body.

18. The adaptor of claim 11 wherein the buffer tube receiver includes a swivel socket.

19. The adaptor of claim 11 wherein the buffer tube receiver includes a rail interface for detachably connecting firearm accessories to the adaptor.

20. The adaptor of claim 19 wherein the buffer tube receiver includes a second rail interface for detachably connecting the firearm stock to the adaptor.

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